

You have been hired to manage a newly established personal care products company with an initial balance sheet of €66 million. The company can produce Sunscreen (S) and/or Moisturizer (H) and sell in three markets: two domestic markets (A and B) and one foreign market (E). Only the companies participating in the simulation operate in markets A and B. In market E, your company will also compete with three local companies. You will manage the company over several years, and your goal is to achieve a higher accumulated profit than your competitors. Follow these steps:

1. Analyze your resources and the market, and set your sales target

1.1. Resources

Your company's initial balance sheet is as follows:

ASSETS		EQUITY	
Fixed assets	12.000.000	Capital	66.000.000
Inventory	-		
Cash and cash equivalents	54.000.000		

Fixed assets include land and buildings.
Cash and cash equivalents are the funds the company has available to begin producing and selling. Shareholders have contributed €66 million in capital.

1.2. Demand

You must determine your sales target. Begin by calculating the overall demand for each product in each market using the following information:

Market	Area (km²)	Population	Per capita consumption (units)	
			S	H
A	100.000	22.000.000	0,55	1,33
B	95.000	18.000.000	0,50	1,50
E	210.000	30.000.000	1,17	2,50



These are frequently purchased consumer products.
Demand figures may be affected by per capita income and marketing efforts.
Based on this overall demand, decide the market share you want to achieve and the units of each product you want to sell in each market.

1.3. Distribution channels

In markets A and B, your company can sell its products through 3 retail channels, while market E can be accessed through a single channel. The following table shows the total number of points of sale in each channel:

Channel		A	B	E
G	Hypermarkets	35	30	200
S	Supermarkets	1.620	1.080	
T	Traditional stores	2.160	1.080	

1.4. Commercial distribution and logistics

Markets A and B each have 6 sales areas, while market E has 12.
In each area, decide whether your company will sell to the retail channel through an internal sales force (sales managers) or independent agents (wholesalers). Once a sale is made, the product is transported from the factory (F) to a platform (P), and from there to the point of sale (D).
The sales and logistics costs associated with sales managers/wholesalers are as follows:

	Fixed expenses (€)	Variable costs (%)	Logistics costs
	Sales managers €50,000 per manager in markets A and B. \$60,000 per manager in E. Sales reps salaries.	Optional variable compensation for sales reps in A and B.	Transportation from Factory to Platform and from Platform to Retailer.* In E, a fixed cost of €0.024 per unit for transportation to the main platform. Administrative costs of €0.65 + €0.15/\$ per order. Handling costs of €0.0027/\$ per unit sold. Storage costs of €2/\$ per liter stored on the platform. Tariffs in market E: 2% of production cost + €0.024.
Wholesalers		8.46% (regular units). 6.5% (promotional units).	Transportation from Factory to Platform.* In E, a fixed cost of €0.024 per unit for transportation to the main platform.

* Cost per truck: €24,000 + 1.2 × load capacity in kilograms.

1.5. Production facilities and capacity

Fixed assets include the land and buildings required to produce both products.
To produce, you must install and activate separate production lines for S and H. Each year you may install new lines, up to a combined maximum of 32 lines. The investment required to install production lines is €2.1 million for S and €1.5 million for H.
The theoretical production capacity of one S line is 240,000 units per year per shift. For H, it is 600,000 units per year and shift.
Each line operates 2,000 hours per year.
Each year, choose from 1 to 3 shifts. The decision applies to all active lines.

1.6. Maintenance and breakdowns

The theoretical production capacity of both S and H lines will be reduced by two factors: cleaning/maintenance and breakdowns.
A line must undergo maintenance every 40 production hours, stopping the line for 5 hours and causing a corresponding loss of productivity.
For every 2,000 operating hours on a line (one year), 70 breakdowns occur. This number increases proportionally when two or three shifts are used.
Each breakdown causes the loss of 2.5 production hours (0.5 for detection, 0.5 for preparation and 1.5 for repair), in addition to product loss.
Breakdown repairs require 4 external experts who are not on staff, at a cost of €24.041 per hour per expert. These wages are included in standard production cost.

1.7. Productivity improvement measures

You have three ways to reduce the productivity loss caused by maintenance and breakdowns.

Preventive maintenance	Allows 70% of breakdowns to be anticipated, reducing repair time and some product loss. The remaining 30% are still unexpected. This requires hiring specialized engineers at an annual cost of €500,000.
POKA YOKE	This quality control system helps locate a breakdown, saving detection time and preventing product loss. It is an asset requiring an investment of €720,000 per installed line. If implemented, it will apply to all current and future lines.
SMED	This is an exact replica of the main production module, allowing production lines to continue operating during maintenance. As an asset, installation requires an investment of €3 million per replica, which can serve both S and H lines. One SMED unit can be used for several lines, but a number of specialist teams (4 people per team) must be assigned to operate the SMED units.

1.8. Procurement

Each finished unit of S and H requires one unit of raw material, including all necessary ingredients and consumables.
These raw materials initially cost €1.65 per unit for S and €1.05 per unit for H. The costs will be affected by purchase volume and payment terms.
Raw material for one product cannot be used to manufacture the other; the materials are not interchangeable.
If you do not purchase enough raw material to meet production needs, the simulator will purchase the shortfall at a 15% higher cost.

1.9. Standard production cost

Standard production cost is the sum of raw material cost, production personnel wage, and energy cost.
Energy costs €0.25 per unit of S produced and €0.15 per unit of H.

1.10. Human Resources

The company has three types of employees: office and management staff, production personnel and sales staff. The following table provides details for each type.

Area	Type	Number of employees	Minimum base salary	Maximum base salary	Minimum variable pay	Maximum variable pay
Office and management	General management	1	108.000	150.000	N/A	N/A
	Department managers	8	72.124	99.170	N/A	N/A
	Tactical decisions by function	15	51.100	72.124	N/A	N/A
	Executors	32	30.050	42.075	N/A	N/A
Production	Specialists	9 (S) and 6 (H)	21.000	29.450	N/A	N/A
	Operators	32 (S) and 23 (H)	12.000	16.830	N/A	N/A
Sales	Market A (supermarkets and traditional stores)	To be decided	21.000	42.000	0% or 0.6%	1,2%
	Market B (supermarkets and traditional stores)	To be decided	21.000	42.000	0% or 0.6%	1,2%

Low salaries will lead to turnover (employees leaving the company) among office and sales staff, and absenteeism (lower productivity) among production staff. This will result in replacement costs for departing office and management and sales staff and a higher unit manufacturing cost.
To reduce the impact, a recruitment and training department may be deployed at an annual cost of €722,000; this decision can be made each year. The unemployment rate will also affect turnover and absenteeism.

1.11. Financial instruments

Your company has the following financing instruments available:

- Factoring: initially subject to an interest rate of 5.5%. Customer payment terms may be 60, 90 or 120 days.
- Supplier payment terms: cash, 60, 90, or 120 days.
- Credit lines: a 1% upfront fee, a 1% service fee and a 7% interest rate.
- Loans: initial term of 4 years (which decreases each year), a 1% upfront fee, a 1% prepayment penalty and a 10% interest rate.

The company also has several financial investment instruments:
1. Financial investments in the form of deposits: initially earn 2.5% interest and carry a 1% early withdrawal fee.
2. Fixed-term deposits: initially earn 2.5% interest and carry a 1% early withdrawal fee.
3. Purchase of company shares: if enabled by the simulation director, prices will be available in the simulator interface.

2. Design an annual business plan to sell your products and generate profit

2.1. Marketing

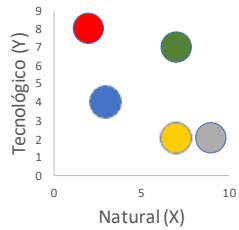
Your first step is to set a market-share and sales target. To achieve it, you must make marketing decisions that generate demand.

Product Decide whether to sell one product or both, and in which markets and channels.

Distribution Decide the number of sales managers to hire in each market: 0 to 6 in A and B, and 0 to 12 in E.
If you do not choose the maximum, independent wholesalers will serve the remaining areas.

Advertising To obtain your brand positioning, select one of the 20 advertising campaigns for each product you will find in the scenario appendix.
For advertising purposes, assume buyers represent 25% of the total population, split equally between women and men.
Prepare a media plan, deciding the number of insertions (appearances) to run in each medium during the year.

This chart shows the positioning of the companies:



This table shows the available advertising media, the cost of one insertion and their reach:

Medium	A and B		E	
	Cost (€)	Reach	Cost (\$)	Reach
Display network (DR)	10.000	28	11.000	26
Press (PR)	15.000	48	16.000	31
Radio (RA)	14.000	35	16.000	40
Social media (SM)	15.000	30	15.000	29
Television (TV)	18.000	60	20.000	61

Reach is the percentage of each buyer type reached by an advertisement.
Ideally, use a media mix to reach 100% of the audience more often than the competition.
There are three consumer segments: 40% seek X products, 40% seek Y products, and 20% seek products that are neither X nor Y.

Prices and margins Set the final retail price. If you do not enter a price for a channel, the simulator will assume you do not want to sell through that channel.
You must also decide the margin, in euros, to give the retailer for selling your products.

Sales promotions Your company may run up to 3 promotions per year and channel, always of the same type within that channel.
There are eight types of promotions: price reduction, 3-for-2, discount on the next purchase, raffle, one direct promotional bonus and 3 indirect bonuses.
For each channel, decide whether to run promotions, how many to run, the type, and the specific terms (detail).

Shelf space Shelf space gives the product a presence at the point of sale and therefore helps increase sales. It is the amount of display space occupied by the product.
Choose a shelf space percentage consistent with your sales target.

Point-of-sale advertising For each retail channel in which you want to sell, set a budget in euros for point-of-sale advertising.
Point-of-sale advertising includes posters, headers, and other in-store promotional materials. The maximum is €500,000 per channel per year.



Warning! If you do not enter prices, margins, and shelf space for a channel, the simulator will assume you do not want to sell through that channel.
Dumping is not allowed; you cannot sell below production cost plus the retailer margin.

Market research The reports provide data on your competitors' actions and their results. They are essential for preparing a new plan.
Not commissioning market research to save money is a bad idea.

2.2. Production

Based on your sales target, decide how many S and H lines to install and activate. Remember that theoretical and actual capacity differ.
Choose the number of shifts and the amount of raw material to purchase. Raw material inventory from one year may be used the following year.
To improve productivity, you can implement preventive maintenance, Poka Yoke and SMED (from year 2 onward).
If you install SMED, you must also decide how many special teams you will hire to operate them.

2.3. Human Resources

Decide the salaries of office and management staff and production employees, taking the workplace climate into account.
If you use managers, they will handle sales to channel G. For channels S and T, decide how many sales reps to hire per area (not per market) and set their salaries.

2.4. Finance

Compare your available cash with the Operating Cash Required (OCR), consisting of:
- 80% of investment in fixed assets,
- 50% of the sum of advertising expenses, sales managers expenses and sales reps fixed salaries, overhead expenses, and market research,
- and 15% of raw material cost.

If your OCR exceeds your available cash, you must take a loan.
You will also decide supplier payment terms, customer collection terms, whether do factoring and the maximum amount, and whether to take a credit line.
Finally, if you have excess cash after completing your financial plan, you may make a financial investment, open a fixed-term deposit, or buy stocks.



Warning! The repayment term for new loans decreases each simulation year.
If your calculation is incorrect, the simulator will request the amount you actually need to avoid beginning the year with an overdraft.
You may take a credit line as a buffer in case your plan is not achieved, thereby avoiding a 20% overdraft interest.

3. Financial forecast and control

3.1. Revenue to consider

Sales revenue Company revenue equals the retail price multiplied by the number of units sold.
Revenue is shown separately for S and H.

Financial income If you have excess cash, you may invest in a deposit, a fixed-term deposit, or stocks.
Returns on these financial investments are recognized as income.

3.2. Expenses to consider

Retailer margin This is one of the marketing decisions and is deducted from S and H sales revenue.
Total sales less retailer margins equal the company's net revenue.

Direct production cost Direct production cost equals unit cost (raw materials including lost units + salaries including SMED and repair teams + power/energy) multiplied by the number of units sold. It is shown separately for S and H.

Cost of sales promotions These equal the promotion cost multiplied by the number of units sold on promotion.
Highly effective promotions can entail a significant cost.

Wholesalers If independent wholesalers are used, they will charge 8.46% per regular unit sold and 6.5% per promotional unit.

Logistics 1. Factory-to-platform transport expenses.
2. Storage expenses.
3. Platform-to-retailer transport expenses.

Tariffs Import tariffs in market E are 2% of the sum of production cost and transportation to the main platform.

Depreciation Depreciation is the loss in value of an asset over time relative to the investment made.
Production lines, Poka Yoke and SMED are depreciated at 8%, 10% or 12% per year depending on the number of production shifts.

Overhead expense Overhead expenses are fixed costs required for the company to operate.
These amount to €600,000 per year.

R&D The company must continually invest in innovation. This equals 4.5% of sales each year.

Sales network Includes sales managers expense and the fixed portion of sales reps salaries.
They are shown separately by market.

Sales force incentives These are the variable portion of sales reps salaries.
They are calculated as a percentage of net revenue (the selling price to the retailer).

Research expenses Market research is outsourced.
The total budget spent is reported on the income statement.

Advertising expense Includes the total advertising budget invested in media and point-of-sale advertising (PoS).
It is shown separately by product.

Office and management These are office and management staff salaries.
Includes the salaries of all employees according to the salary levels selected.

Training department This is the annual expense of the recruiting and training department.
This will affect employee turnover.

Turnover Expenses caused by office and office and management staff and salespeople leaving the company.

Financial expenses 1. Upfront fees for loans and credit lines, prepayment penalties for loans and early withdrawal fees for investments.
2. Interest on loans, credit lines and factoring, losses on share sales, or overdraft interest.



Generating sales (demand) is very important, but sales must generate profit. Understanding, estimating and controlling revenues and expenses is essential. Try to project your P&L to fine tune your business plan.

PRAXISMMT Corp.